

1. Administrative & Study Identification

Full Study Title: A Phase II, Open-label, Multicentre, Randomised Study of Neoadjuvant and Adjuvant Treatment in Patients With Resectable, Early-stage (II to IIIB) Non-small Cell Lung Cancer (NeoCOAST-2) **Short Title/Acronym:** NeoCOAST-2 **Protocol Number:** [Internal Study ID / AstraZeneca Protocol] **NCT Number:** NCT05061550 **Posting ID:** 521499630002922265 **Sponsor/Company Name:** AstraZeneca **Investigational Product:** Durvalumab combined with novel immunotherapies (oleclumab, monalizumab, volrustomig, AZD0171, rilvegostomig), Datopotamab deruxtecan (Dato-DXd), and platinum chemotherapy (e.g., cisplatin/Platinol) **Phase of Development:** Phase II **Trial Status:** Recruiting **Medical Monitor:** Marina Chiara Garassino (Lead Physician) / AstraZeneca Clinical Operations

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the pathological complete response (pCR) rate (per blinded independent pathologist review) and the safety (adverse events and serious adverse events) of multiple novel immunotherapy, targeted ADC, and chemotherapy combinations. **Secondary Objectives:** To evaluate investigator-assessed event-free survival (EFS), disease-free survival (DFS), overall survival (OS), feasibility to surgery (number of participants having surgical resection), major pathological response (MPR) rate, objective response rate (ORR) following neoadjuvant therapy, pharmacokinetics, immunogenicity, and changes in circulating tumor DNA.

2.2 Background & Rationale Neoadjuvant platinum-based chemotherapy plus immuno-oncology (IO) therapy has been shown to prolong event-free survival and increase pCR rates in patients with resectable NSCLC. The NeoCOAST-2 study builds on this by evaluating multiple novel regimens (targeting pathways like CD73, NKG2A, PD-1/CTLA-4, TROP2) to determine if these combinations can further improve pCR and long-term survival outcomes for patients with early-stage, resectable NSCLC prior to and following surgical resection.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Multi-arm platform (7 experimental arms) **Blinding:** Open-label **Randomization:** Randomized

3.2 Planned Enrollment & Duration Targeted Enrollment: 630 participants **Number of Sites:** Multicenter (across the US, Europe, Canada, and Asia) **Trial Status:** Recruiting **Estimated Primary Completion:** Ongoing/Active

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Newly diagnosed NSCLC patients with histologically/cytologically confirmed resectable disease (Stage IIA to Stage IIIB). 2. WHO or Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1. 3. Adequate organ and bone marrow function. 4. Adequate pulmonary function. 5. Provision of tumor samples (newly acquired or archival tumor tissue $\leq$ 6 months old) to confirm Programmed death-ligand 1 (PD-L1) status, epidermal growth factor receptor (EGFR), or anaplastic lymphoma kinase (ALK) status.

4.2 Exclusion Criteria 1. Participants with sensitizing EGFR mutations or ALK translocations. 2. Participants with baseline PD-L1 expression status $<1\%$ (Applies to Arms 6 and 7 only). 3. Active or prior documented autoimmune or inflammatory disorders; history of active primary immunodeficiency. 4. Uncontrolled intercurrent illness, uncontrolled hypertension, unstable angina pectoris, uncontrolled cardiac arrhythmia, active bleeding diseases, serious chronic gastrointestinal conditions associated with diarrhea, or psychiatric illness/social situations that would limit compliance. 5. History of another primary malignancy. 6. Participants with small-cell lung cancer (SCLC) or mixed small-cell lung cancer. 7. History of non-infectious interstitial lung disease (ILD)/pneumonitis that required steroids, current ILD/pneumonitis, or suspected ILD/pneumonitis that cannot be ruled out by imaging. 8. Preoperative radiotherapy treatment as part of their care plan. 9. Participants who require or may require pneumonectomy, segmentectomies, or wedge resections, as assessed by their surgeon at baseline, to obtain potentially curative resection. 10. QTcF interval $\geq$ 470 ms or moderate/severe cardiovascular disease. 11. Any medical contraindication to treatment with chemotherapy as listed in the local labeling. 12. Any concurrent chemotherapy, investigational product, biologic, or hormonal therapy for cancer treatment. 13. Receipt of live attenuated vaccine within 30 days prior to the first dose of study interventions. 14. Current or prior use of immunosuppressive medication within 14 days before the first dose. 15. Prior exposure to approved or investigational immune-mediated therapy (anti-CTLA-4, anti-TIGIT, anti-PD-1, anti-PD-L1, anti-PD-L2, adenosine pathway, anti-NKG2A, anti-HLA-E, anti-LIF) or previous treatment with a TROP2 targeting ADC or another ADC containing a TOP1 inhibitor. 16. Active or uncontrolled infections including HBA, HBV, HCV, and HIV.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Eligible participants will be randomized to one of the following treatment regimens: * **Arm 1:** Oleclumab + durvalumab + platinum doublet chemotherapy (CTX) (neoadjuvant) $\rightarrow$ Oleclumab + durvalumab (adjuvant). * **Arm 2:** Monalizumab + durvalumab + CTX (neoadjuvant) $\rightarrow$ Monalizumab + durvalumab (adjuvant). * **Arm 3:** Volrustomig (Dose Exploration) + CTX (neoadjuvant) $\rightarrow$ Volrustomig (adjuvant). * **Arm 4:** Datopotamab deruxtecan (Dato-DXd) + durvalumab + single agent platinum chemotherapy (e.g., cisplatin/Platinol) (neoadjuvant) $\rightarrow$ durvalumab (adjuvant). * **Arm 5:** AZD0171 + durvalumab + CTX (neoadjuvant) $\rightarrow$ AZD0171 + durvalumab (adjuvant). * **Arm 6:** Rilvegostomig + CTX (neoadjuvant) $\rightarrow$ Rilvegostomig (adjuvant). * **Arm 7:** Dato-DXd + Rilvegostomig + single agent platinum chemotherapy (neoadjuvant) $\rightarrow$ Rilvegostomig (adjuvant).

Route of Administration: Intravenous (IV) **Duration of Treatment:** Neoadjuvant phase followed by surgical resection, then an Adjuvant phase.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for chemotherapy-induced nausea/vomiting. **Prohibited:** Concurrent systemic anti-cancer therapies, live attenuated vaccines within 30 days, prior exposure to related IO therapies or ADCs, and immunosuppressive medications within 14 days.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	-28 to 0	Screening: Informed Consent, Medical History, PD-L1/EGFR/ALK testing, Staging Scans, Pulmonary/Organ Function evaluation
2-12	Weeks 1-12	Neoadjuvant: Randomization to Arms 1-7, IV Infusions, Safety Labs, Tumor Assessment
3	Within 40 days	Surgery: Surgical resection (excluding pneumonectomy/segmentectomy/wedge), Pathology review
4	Start within 10 wks	Adjuvant: Initiation of specific adjuvant IO or targeted therapy based on Arm, Efficacy & Safety monitoring
5	Up to 5+ years	End of Study: Survival Follow-Up, DFS/OS tracking

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Efficacy:** Number of participants with pathological complete response (pCR) via blinded independent pathologist review. * **Safety:** Number of participants with adverse events (AEs) and serious adverse events (SAEs).

7.2 Secondary & Exploratory Endpoints * Number of participants experiencing an event-free survival (EFS) event. * Number of participants experiencing a disease-free survival (DFS) event. * Number of participants having surgical resection. * Major Pathological Response (MPR) rate and Objective Response Rate (ORR). * Pharmacokinetics, immunogenicity, and changes in circulating tumor DNA (ctDNA).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of AEs graded by CTCAE, focused heavily on immune-related adverse events (irAEs), ILD/pneumonitis, and surgical feasibility/delays. **Serious Adverse Events (SAEs):** Standard 24-hour reporting timelines. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee assigned to review safety across all 7 experimental multi-arms.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for all efficacy measures (pCR, EFS, OS). Safety Analysis Set for all patients receiving at least one dose of the study drug. **Sample Size Justification:** Target enrollment of 630 participants powered to assess definitive signals in pCR improvements and safety profiles across the experimental arms compared to benchmarks. **Handling of Missing Data:** Patients who do not undergo surgery or lack pathological evaluation are considered non-responders for pCR/MPR endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring for data validation, focusing on adherence to neoadjuvant dosing schedules and surgical windows. **Audit Trail:** eCRF locking and independent blinded pathology review ensuring unbiased primary endpoint calculations.

11. Study Identifiers & Governance

Primary ID: NCT05061550 **Registry Number:** NCT05061550 **Posting ID:** 521499630002922265 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** NeoCOAST-2 **Official Regulatory Title:** A Phase II, Open-label, Multicentre, Randomised Study of Neoadjuvant and Adjuvant Treatment in Patients With Resectable, Early-stage (II to IIIB) Non-small Cell Lung Cancer

12. Clinical Framework (Oncology / NSCLC Specific)

Primary Condition: Early-stage (IIA to IIIB) Non-small Cell Lung Cancer (NSCLC) **Diagnosis Threshold:** Histologically/cytologically confirmed resectable disease without EGFR/ALK mutations; Stage IIA-IIIB. **Medical Methodology:** Perioperative Therapy (Neoadjuvant targeted ADC/IO+IO+CT followed by Surgery and Adjuvant targeted ADC/IO) **Optimization Target:** Pathological Complete Response (pCR), Event-Free Survival (EFS), and surgical feasibility.

13. Study Procedures & Methodology

Management Pathway: Multi-disciplinary Perioperative Oncology Clinical Pathway **Education Protocol (HCP):** Site initiation visits detailing novel IO combinations (including TROP2 ADC Dato-DXd), toxicity management (especially ILD/pneumonitis), and strict surgical timing parameters. **Education Protocol (Patient):** Informed consent outlining multi-stage treatment (chemo-immunotherapy, surgery, post-operative immunotherapy). **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) of pathology and investigator tracking of feasibility to surgery.

14. Observation & Follow-Up Schedule

Total Duration: Treatment up to ~1.3 years; Survival Follow-up up to 5+ years **Onsite Visit Cadence:** Every 3 weeks during Neoadjuvant and Adjuvant treatment phases **Remote Monitoring Frequency:** Telephone safety follow-ups as determined by specific cycle requirements. **Checklist Review Interval:** Routine restaging imaging (CT/MRI) according to institutional and protocol standards to monitor Disease-Free Survival (DFS) and Event-Free Survival (EFS).

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	Marina Chiara Garassino	____	03-Apr-2026	Clinical Operations Lead	AstraZeneca		
Representative	____	03-Apr-2026	Lead Statistician	[Name]			
	____	03-Apr-2026					